

Date	July 1, 2026
Team ID	SB-AI-2026-9284
Project Name	Sustainable Waste Management AI Assistant
Maximum Marks	2 Marks

Scalability & Future Roadmaps

1. Advanced AI Integrations

- ? Browser Edge Object Recognition: Load lightweight TensorFlow.js models directly on the client camera to eliminate network latency for common objects.
- ? Multi-modal Analysis: Allow direct image uploads to Llama 3.2 vision models to parse complex packages directly.

2. Municipal Systems Connectors

- ? Smart dumpster logging: Integrate API feeds from weight and volume sensors inside city sorting dumps.
- ? PWA Offline access: Setup service workers to cache Leaflet maps, enabling residents to check instructions in basements or offline regions.